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Optical fiber cable with partitioned case for optical fiber cable

Abstract

An optical fiber cable includes a plurality of first optical fiber cords including a trunk optical fiber cord and a branch optical fiber cord, a second optical fiber cord connected to the branch optical fiber cord, and a case covering a part of the plurality of first optical fiber cords and a part of the second optical fiber cord, where the case includes a first tube portion through which a trunk optical fiber cord is passed, a second tube portion adjacent to the first tube portion and through which a branch optical fiber cord is passed, and a partition wall partitioning the first tube portion and the second tube portion.

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Background/Summary

TECHNICAL FIELD

- (1) The present disclosure relates to an optical fiber cable and a case for an optical fiber cable.
- (2) The present application claims priority to Japanese Patent Application No. 2021-197035 filed on Dec. 3, 2021, the entire contents of which are incorporated herein by reference.

BACKGROUND

(3) Japanese Unexamined Patent Publication Nos. 2001-51131, 2001-116968, and 2001-201641, and U.S. Pat. Nos. 7,127,143, 7,346,243, 7,693,374, 8,582,938, 10,437,003, and 10,712,519 each disclose various optical fiber cables used in optical communication. Japanese Unexamined Patent Publication No. 2005-208193 discloses an optical microbox for connecting a branch cable branched from an optical trunk cable and a termination cable. U.S. Pat. No. 10,371,917 discloses a data center comprising an optical fiber or the like.

SUMMARY

(4) An optical fiber cable according to an aspect of the present disclosure includes a plurality of first optical fiber cords including a trunk optical fiber cord and a branch optical fiber cord, a second optical fiber cord connected to the branch optical fiber cord, and a case covering a portion of the plurality of first optical fiber cords and a portion of the second optical fiber cord. The case includes a first tube portion through which a trunk optical fiber cord is passed, a second tube portion adjacent to the first tube portion and through which a branch optical fiber cord is passed, and a partition wall partitioning the first tube portion and the second tube portion.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic plan view illustrating an optical communication system including an optical fiber cable according to an embodiment.
- (2) FIG. 2 is a schematic plan view illustrating a main part of the optical fiber cable shown in FIG. 1.
- (3) FIG. 3 is a schematic cross sectional view illustrating a part of an optical fiber cable.
- (4) FIG. 4A is a schematic perspective view illustrating a branch protection case, and FIG. 4B is a schematic side view illustrating the branch protection case.
- (5) FIG. 5 is a schematic perspective view illustrating an open state of the branch protection case.
- (6) FIG. 6 is a schematic view illustrating a main part of an optical fiber cable to which a branch protection case is attached.
- (7) FIG. 7 is a schematic perspective view illustrating a branch protection case according to a modification.

DETAILED DESCRIPTION

Problems to be Solved by the Present Disclosure

(8) In a data center or the like as described in U.S. Pat. No. 10,371,917, for example, an optical cable including a plurality of optical fiber cords (tethers) as shown in U.S. Pat. No. 10,437,003 is used. If necessary, an optical fiber cord is branched from the optical cable. A branched optical fiber cord (especially, a position branched from an optical cable, or the like) is likely to be easily damaged than the trunk line. Therefore, there is a demand for an optical fiber cable and a case for an optical fiber cable capable of suppressing damage to an optical fiber cord branched from a trunk line.

Effects of the Present Disclosure

(9) According to the present disclosure, it is possible to provide an optical fiber cable and an optical fiber cable case capable of suppressing damage to an optical fiber cord branched from a trunk line.

Description of Embodiments of the Present Disclosure

(10) First, contents of embodiments of the present disclosure will be listed and described.

(11) An embodiment of the present disclosure is an optical fiber cable including: a plurality of first optical fiber cords including a trunk optical fiber cord and a branch optical fiber cord; a second optical fiber cord connected to the branch optical fiber cord; and a case covering a part of the plurality of first optical fiber cords and a part of the second optical fiber cord, where the case includes: a first tube portion through which the trunk optical fiber cord is passed; a second tube portion adjacent to the first tube portion and through which the branch optical fiber cord is passed; and a partition wall partitioning the first tube portion and the second tube portion, wherein an optical connecting portion of the branch optical fiber cord and the second optical fiber cord is accommodated in the second tube portion, a first end of the second tube portion is located between a first end of the first tube portion and a second end of the first tube portion, the first end of the second tube portion communicating with an interior space of the first tube portion, and the partition wall extends from the first end of the second tube portion to a second end of the second tube

portion.

(12) According to this optical fiber cable, since the partition wall extending from the first end of the second tube portion to the second end of the second tube portion is provided, the trunk optical fiber cord and the branch optical fiber cord can be separated from each other in the case. Thus, interference between the trunk optical fiber cord and the branch optical fiber cord in the case can be suppressed. In addition, since the first end of the second tube portion communicates with the interior space of the first tube portion, a branch start position of the branch optical fiber cord can be covered by the case. Accordingly, since the branch start position can be favorably protected by the case, it is possible to suppress damage to the branch optical fiber cord branched from the trunk optical fiber cord.

(13) The optical fiber cable may further include: a first tube located at least outside the case, the first tube covering the plurality of first optical fiber cords; a second tube located at least outside the case, the second tube covering the second optical fiber cord; a first resin member surrounding at least the first end of the first tube portion; and a second resin member surrounding at least the second end of the first tube portion and the second end of the second tube portion, where the first resin member may be in contact with at least a first part of the case and a part of the first tube, and the second resin member may be in contact with at least a second part of the first tube, and a part of the second tube. In this case, the resin member can suppress rattling of the case in the optical fiber cable.

(14) Each of the first resin member and the second resin member may have a thermal shrinkage property.

(15) The case may further include a body including the first tube portion, the second tube portion, and the partition wall, and the body may be an assembly structure of a first main body portion and a second main body portion, the first main body portion defining a first part of the first tube portion, a first part of the second tube portion, and a first part of the partition wall, and the second main body portion defining a second part of the first tube portion, a second part of the second tube portion, and a second part of the partition wall. In this case, the case can easily accommodate the optical fiber cable and the optical fiber cord.

(16) The case may further include a connecting portion connecting the first main body portion and the second main body portion. In this case, the first main body portion and the second main body portion can be easily assembled.

(17) The optical fiber cable may further include a sleeve covering the optical connecting portion, and an outer face of the sleeve may be sandwiched by the second tube portion. In this case, since a movement of the optical connecting portion in the second tube portion can be suppressed, the optical connecting portion is less likely to be damaged.

(18) A projection located between the first end and the second end of the second tube portion may be provided on an inner wall of the second tube portion, and the projection may be in contact with the outer face. In addition, the projection may have elasticity. In this case, it is possible to suppress damage to the part of the sleeve that is in contact with the projection.

(19) An embodiment of the present disclosure is a case for an optical fiber cable, the case including: a first tube portion extending in a first direction; a second tube portion extending in the first direction, the second tube portion being adjacent to the first tube portion in a second direction orthogonal to the first direction; and a partition wall partitioning the first tube portion and the second tube portion, where an inner diameter of the first tube portion is larger than an inner diameter of the second tube portion, a first end of the second tube portion is located between a first end of the first tube and a second end of the first tube portion in the first direction, the first end of the second tube portion communicating with an interior space of the first tube portion, a second end of the second tube portion is located closer to the second end of the first tube portion than the first end of the second tube portion in the first direction, and the partition wall extends from the first end of the second tube portion to the second end of the second tube portion in the first direction.

(20) According to this case, since the inner diameter of the first tube portion is larger than the inner diameter of the second tube portion, for example, the first tube portion can accommodate an optical fiber cable as a trunk line, and the second tube portion can accommodate an optical fiber cord branched from the optical fiber cable. Here, since the partition wall extending from the first end of the second tube portion to the second end of the second tube portion in the first direction is provided, the optical fiber cable and the optical fiber cord can be separated from each other in the case. Thus, it is possible to suppress interference between the optical fiber cable and the optical fiber cord in the case. In addition, since the first end of the second tube portion communicates with the interior space of the first tube portion, the branch start position of the optical fiber cord can be covered by the case. Accordingly, since the branch start position can be favorably protected by the case, it is possible to suppress breakage of the optical fiber cord branched from the optical fiber cable.

(21) A projection located between the first end and the second end of the second tube portion in the first direction may be provided on an inner wall of the second tube portion. In this case, since the optical fiber cord in the second tube portion is in contact with the projection, the movement of the optical fiber cord in the second tube portion can be suppressed.

(22) The projection may be elastic. In this case, damage to a part of the optical fiber cord in contact with the projection can be suppressed.

Details of Embodiment of the Present Disclosure

(23) Specific examples of an optical fiber cable and a case for an optical fiber cable according to an embodiment of the present disclosure will be described below with reference to the drawings. It should be noted that the present disclosure is not limited to these examples, but is defined by the scope of claims and is intended to include all modifications within the meaning and scope equivalent to the scope of claims. In the following description, the same reference numerals are given to the same elements in the description of the drawings, and redundant description will be omitted.

(24) FIG. 1 is a schematic plan view illustrating an optical communication system including an optical fiber cable according to the present embodiment. An optical communication system **100** illustrated in FIG. 1 is a group of devices provided in a data center, a base station, or the like, and includes a server rack group **101**, **102**, a distribution frame **103**, **104**, and optical fiber cables **1** to **4**.

(25) Each server rack group **101**, **102** is formed by arranging n server racks (n is a natural number of 2 or more) in order in a predetermined direction. The server rack group **101** includes a plurality of server racks **101-1** to **101- n** arranged in one direction in a plan view. Each of the server racks **101-1** to **101- n** is a rack on which a plurality of physical servers (not illustrated) are placed. Each of the server racks **101-1** to **101- n** is provided with a plurality of shelves (not shown) arranged in a vertical direction. A physical server or the like is placed on each of the plurality of shelves. Like the server rack group **101**, the server rack group **102** includes a plurality of server racks **102-1** to **102- n** arranged in one direction in a plan view.

(26) Each of the distribution frames **103**, **104** is a line concentrator that accommodates a communication line (optical fiber cable) used in the optical communication system **100**. External multi-core cables (not shown) connected to each of the distribution frames **103**, **104** are respectively divided and arranged into a plurality of communication lines. In the distribution frames **103**, **104**, termination processing of optical fiber cables may be performed. A termination unit that holds the multi-core optical wiring and accommodates connection points may be mounted on the distribution frames **103**, **104**. In the present embodiment, each of the distribution frames **103**, **104** is an intermediate distribution frame (IDF), but is not limited thereto. In addition, the distribution frame **103** is connected to an external multi-core cable (not illustrated) which is a working line, and the distribution frame **104** is connected to an external multi-core cable (not illustrated) which is a spare line.

(27) The optical fiber cable **1** is a multi-core cable connecting the server rack group **101** and the

distribution frame **103**. Similarly, the optical fiber cable **2** is a multi-core cable connecting the server rack group **101** and the distribution frame **104**, the optical fiber cable **3** is a multi-core cable connecting the server rack group **102** and the distribution frame **103**, and the optical fiber cable **4** is a multi-core cable connecting the server rack group **102** and the distribution frame **104**. The optical fiber cables **1** and **3** are used as working lines, and the optical fiber cables **2** and **4** are used as protection lines. The optical fiber cables **1** to **4** may have the same structure or may have different structures.

(28) FIG. **2** is a schematic plan view illustrating the optical fiber cable **1** shown in FIG. **1**. As illustrated in FIG. **2**, the optical fiber cable **1** includes a trunk cable **10**, multicore optical fiber cords **11** to **14**, branch optical fiber cables **15-1** to **15-n** (n is a natural number of 2 or more), connectors **21** to **24**, a branch member **31**, and branch protection cases **40-1** to **40-n**. Here, the branch optical fiber cables **15-1** to **15-n** have the same structure each other, and the branch protection cases **40-1** to **40-n** have the same structure each other. Therefore, hereinafter, the details of the optical fiber cable **15-1** among the branch optical fiber cables **15-1** to **15-n** and the details of the branch protection case **40-1** among the branch protection cases **40-1** to **40-n** will be described.

(29) FIG. **3** is a schematic diagram illustrating a main part of an optical fiber cable. In FIG. **3**, the branch protection case **40-1** is omitted. As shown in FIG. **3**, the trunk cable **10** is a main portion located between the server rack group **101** and the distribution frame **103** in an extending direction of the optical fiber cable **1**. One end of the trunk cable **10** is accommodated in the branch member **31**. The trunk cable **10** includes a plurality of optical fiber cords **10a** (first optical fiber cords) and a reinforced tube **10b** (a first tube) covering the optical fiber cords **10a**.

(30) Each of the plurality of optical fiber cords **10a** is, for example, a cord including a plurality of core wires. In the present embodiment, the optical fiber cord **10a** is a six core cord, but is not limited thereto. The lengths of the plurality of optical fiber cords **10a** may be predetermined or may be equal to each other. The reinforced tube **10b** is a member that bundles and protects a plurality of optical fiber cords **10a**. A part of the reinforced tube **10b** in the branch protection case **40-1** is removed. Therefore, the reinforced tube **10b** is located at least outside the branch protection case **40-1**.

(31) Hereinafter, among the plurality of optical fiber cords **10a** in FIG. **3**, the trunk line accommodated in the reinforced tube **10b** is referred to as a trunk optical fiber cord **10c**, and the branch line branching from the trunk optical fiber cord **10c** is referred to as a branch optical fiber cord **10d**. A part of the trunk optical fiber cord **10c** may be exposed from the reinforced tube **10b**. This part is accommodated in the branch protection case **40-1**. The branch optical fiber cord **10d** corresponds to a part of the branch optical fiber cable **15-1**. A core wire **10e** (first core wire) included in the branch optical fiber cord **10d** is exposed at a distal end portion of the branch optical fiber cord **10d**. The distal end portion is accommodated in the branch protection case **40-1**.

(32) Each of the multicore optical fiber cords **11** to **14** is a communication line located between the distribution frame **103** and the branch member **31** in the extending direction of the optical fiber cable **1**. Each of the multicore optical fiber cords **11** to **14** is a cord including more core wires than the optical fiber cord **10a**. The number of core wires included in each of the multicore optical fiber cords **11** to **14** may be equal to a multiplier of the number of optical fiber cords **10a** included in one multicore optical fiber cord and the number of core wires included in the optical fiber cord **10a**. In the present embodiment, each of the multicore optical fiber cords **11** to **14** is a 24-core cord, and thus includes four optical fiber cords **10a**. One end of each of the multicore optical fiber cords **11** to **14** is accommodated in the branch member **31**. The other end of the multicore optical fiber cord **11** is accommodated in the connector **21**. Similarly, the other ends of the multicore optical fiber cords **12** to **14** are accommodated in connectors **22** to **24**, respectively. Each of the multicore optical fiber cords **11** to **14** is reinforced by, for example, a reinforced tube.

(33) Each of the connectors **21** to **24** is an interface connected to the distribution frame **103**. Each of the connectors **21** to **24** is, for example, a multi-fiber push-on (MPO) connector. In this case, the

number of optical fibers coupled to the connector **21** corresponds to the number of optical fibers included in the multicore optical fiber cord **11**. The same applies to the connectors **22** to **24**.

(34) The branch member **31** is a member that accommodates a portion branched from the multicore optical fiber cords **11** to **14** to optical fiber cords **10a**. A cavity is provided in the branch member **31**, and one end of the multicore optical fiber cords **11** to **14** and one end of the optical fiber cords **10a** are accommodated in the cavity. In the branch member **31**, the respective positions of the multicore optical fiber cords **11** to **14** and the respective positions of the optical fiber cords **10a** may be determined. In the branch member **31**, the core wires in the multicore optical fiber cords **11** to **14** are exposed by removing the reinforced tubes or the like provided on the multicore optical fiber cords **11** to **14**. The core wires are bundled as a core wire of each optical fiber cord **10a**. In this case, the branch member **31** also functions as a protective member for the core wire. In addition, since there is no fusion spliced portion between the multicore optical fiber cords **11** to **14** and the optical fiber cords **10a**, optical loss is less likely to occur in the branch member **31**. In the branch member **31**, the positions of each optical fiber cord **10a** and the positions of the multicore optical fiber cords **11** to **14** may be determined. In the branch member **31**, a fusion spliced portion of the multicore optical fiber cord **11** and the optical fiber cords **10a** may be provided. In this case, the branch member **31** also functions as a protection member for the fusion spliced portion or the like.

(35) Each of the branched optical fiber cables **15-1** to **15-n** is a communication line connected to the server rack group **101**. As illustrated in FIG. 3, the branch optical fiber cable **15-1** includes a branch optical fiber cord **10d**, a branch trunk cable **51**, optical fiber cords **52** to **54** (second optical fiber cords), a protection member **55**, connectors **56** to **58**, and a sleeve **59**. The branch trunk cable **51** is a trunk line in the branch optical fiber cable **15-1** and is optically connected to the optical fiber cord **10a**. Therefore, the branch trunk cable **51** before the optical connection is a line separate from the trunk cable **10**. The branch trunk cable **51** has a reinforced tube **51a** (second tube) that accommodates the optical fiber cords **52** to **54**. A part of the reinforced tube **51a** is removed in the branch protection case **40-1**. Therefore, the reinforced tube **51a** is located at least outside the branch protection case **40-1**.

(36) Each of the optical fiber cords **52** to **54** is a cord accommodated in the branch trunk cable **51** and is optically connected to the branch optical fiber cord **10d**. One ends of the optical fiber cords **52** to **54** are accommodated in the branch protection case **40-1**. The other ends of the optical fiber cords **52** to **54** are connected to connectors **56** to **58**, respectively. The optical fiber cord **52** includes a core wire **52a** (second core wire) whose number is lower than number of core wires in the optical fiber cord **10a**, and a tube **52b** covering the core wire **52a**. Each of the optical fiber cords **53**, **54** has the same structure as the optical fiber cord **52**. Therefore, the optical fiber cord **53** includes a core wire **53a** and a tube **53b**, and the optical fiber cord **54** includes a core wire **54a** and a tube **54b**.

(37) A part of the core wire **52a** is exposed from the reinforced tube **51a** in the branch protection case **40-1**. The same applies to core wires **53a** and **54a**. The exposed core wires **52a**, **53a**, and **54a** are respectively fusion spliced to the core wires **10e** in the branch protection case **40-1**. For example, the core wires **52a**, **53a**, and **54a** fixed in alignment with each other and the core wires **10e** fixed in alignment with each other are fusion spliced to each other. The total number of core wires **52a** to **54a** corresponds to the number of core wires included in the branch optical fiber cord **10d**. In the present embodiment, each of the optical fiber cords **52** to **54** includes two core wires. A part of the tube **52b** is exposed between the protection member **55** and the connector **56**. Another part of the tube **52b** is exposed from the reinforced tube **51a** in the branch protection case **40-1**. The same applies to the tubes **53b**, **54b**.

(38) The protection member **55** is a member for protecting a portion branched from the branch trunk cable **51** to the optical fiber cords **52** to **54**.

(39) Each of the connectors **56** to **58** is an interface connected to the server rack group **101**. Each of the connectors **56** to **58** is, for example, a data link connector. The number of optical fibers coupled

to the connector **56** corresponds to the number of optical fibers included in the optical fiber cord **52**. The same applies to the connectors **57**, **58**.

(40) The sleeve **59** is a cylindrical member that protects an optical connecting portion connecting the branch optical fiber cord **10d** and the optical fiber cords **52** to **54**. The sleeve **59** is, for example, a resin member formed so as to cover the optical connecting portion. An outer face **59a** of the sleeve **59** may contact the branch protection case **40-1**. In the present embodiment, the optical connecting portion is a fusion spliced portion.

(41) Next, the structure of the branch protection case **40-1** will be described with reference to FIG. **4A**, FIG. **4B**, FIG. **5** and FIG. **6**. FIG. **4A** is a schematic perspective view illustrating the branch protection case, and FIG. **4B** is a schematic side view illustrating the branch protection case. FIG. **5** is a schematic perspective view illustrating an open state of the branch protection case. FIG. **6** is a schematic view illustrating a main part of an optical fiber cable to which a branch protection case is attached.

(42) As shown in FIGS. **4A**, **4B**, FIG. **5** and FIG. **6**, the branch protection case **40-1** is a member (a case for an optical fiber cable) that protects the branching position of the branched optical fiber cable **15-1** and the portion of the branched optical fiber cable **15-1** where the core wires are exposed. In the present embodiment, the branch protection case **40-1** is also a member that protects the fusion spliced portion between the core wire **10e** and the core wires **52a**, **53a** and **54a**. The branch protection case **40-1** is, for example, a resin molded body. At least a part of the branch protection case **40-1** has elasticity. The branch protection case **40-1** includes a main body **60** having a first tube portion **61**, a second tube portion **62**, and a partition wall **63** partitioning the first tube portion **61** and the second tube portion **62**. In the following description, a longitudinal direction of the first tube portion **61** is referred to as a first direction X, and a direction in which the first tube portion **61** and the second tube portion **62** are arranged is referred to as a second direction Y. The second direction Y coincides with one of the inner diameter directions of the first tube portion **61** orthogonal to the first direction X.

(43) The main body **60** is an assembly structure of a first main body portion **60a** defining a first part **61a** of the first tube portion **61**, a first part **62a** of the second tube portion **62**, and a first part **63a** of the partition wall **63**, and a second main body portion **60b** defining a second part **61b** of the first tube portion **61**, a second part **62b** of the second tube portion **62**, and a second part **63b** of the partition wall **63**. The first main body portion **60a** and the second main body portion **60b** are rotatably connected to each other by connecting portions **60c** and **60d** such as hinges. When most or all of the first main body portion **60a** and the second main body portion **60b** are separated from each other, the branch protection case **40-1** is in an open state. In the open state, a part of the trunk cable **10** and a part of the branch optical fiber cable **15-1** are accommodated in the first main body portion **60a** or the second main body portion **60b**. Thereafter, when the first main body portion **60a** and the second main body portion **60b** are connected each other (connection state), the branch protection case **40-1** is mounted on the trunk cable **10** and the branch optical fiber cable **15-1**. Although not shown, at least one of the first main body portion **60a** and the second main body portion **60b** is provided with a mechanism for maintaining the connection state.

(44) The first tube portion **61** is a portion that protects the trunk line of the optical fiber cable **1**, extends in the first direction X, and has a cylindrical shape. The one end **61c** of the first tube portion **61** in the first direction X is an open end corresponding to one end of the main body **60** in the first direction X. The other end **61d** of the first tube portion **61** in the first direction X is an open end corresponding to the other end of the main body **60** in the first direction X. The inner diameter **D1** of the first tube portion **61** may be equal to or larger than the thickness of the trunk cable **10**. From the viewpoint of preventing rattling between the branch protection case **40-1** and the trunk cable **10** in the connection state, the inner diameter **D1** of the first tube portion **61** is, for example, 1.05 times or less the thickness of the trunk cable **10**. In the connection state, the trunk cable **10** passes through the first tube portion **61**. When at least a part of the trunk cable **10** is accommodated

in the first tube portion **61**, it can be considered that the trunk cable **10** passes through the first tube portion **61**. As described above, a part of the trunk optical fiber cord **10c** exposed from the reinforced tube **10b** is accommodated in the branch protection case **40-1**. Therefore, in the connection state, it can be considered that the trunk optical fiber cord **10c** passes through the first tube portion **61**. In the present embodiment, a reinforced tube **10b** is present at the one end **61c** of the first tube portion **61** and the vicinity thereof, and at the other end **61d** of the first tube portion **61** and the vicinity thereof.

(45) The second tube portion **62** is a portion for protecting a branch line of the optical fiber cable **1**, and is adjacent to the first tube portion **61** in the second direction Y. The second tube portion **62** includes a main tube portion **71** and a connecting tube portion **72** that connects the main tube portion **71** and the first tube portion **61**. The main tube portion **71** is a portion that extends in the first direction X and has a cylindrical shape. The connecting tube portion **72** is a portion for communicating with both an interior space of the main tube portion **71** and the interior space of the first tube portion **61**. Since the connecting tube portion **72** is provided, the interior space of the second tube portion **62** communicates with the interior space of the first tube portion **61**. The one end **62c** of the second tube portion **62** in the first direction X is an open end that is included in the connecting tube portion **72** and communicates with the interior space of the first tube portion **61**. The one end **62c** is located between the one end **61c** and the other end **61d** of the first tube portion **61** in the first direction X. The other end **62d** of the second tube portion **62** in the first direction X is an open end included in the main tube portion **71**, and is located closer to the other end **61d** of the first tube portion **61** than the one end **62c** of the second tube portion **62** in the first direction X. In the present embodiment, in the first direction X, the position of the other end **61d** of the first tube portion **61** and the position of the other end **62d** of the second tube portion **62** are aligned with each other.

(46) The inner diameter D2 of the second tube portion **62** may be equal to or larger than the thickness of the branch optical fiber cable **15-1**. Therefore, the inner diameter D1 of the first tube portion **61** is larger than the inner diameter D2. In the connection state, the branch optical fiber cable **15-1** passes through the second tube portion **62**. A part of the optical fiber cords **52** to **54** included in the branched optical fiber cable **15-1** is exposed from the reinforced tube **51a** in the second tube portion **62**, and the part is accommodated in the second tube portion **62**. Therefore, the optical fiber cords **52** to **54** can be regarded as passing through the second tube portion **62** of the branch protection case **40-1**. As described above, the branch optical fiber cord **10d** is accommodated in the branch protection case **40-1**. Therefore, in the connection state, it can be considered that the branch optical fiber cord **10d** passes through the second tube portion **62**.

(47) In this embodiment, a branch optical fiber cord **10d** is present at the one end **62c** of the second tube portion **62** and the vicinity thereof, and the branch trunk cable **51** is present at the other end **62d** of the second tube portion **62** and the vicinity thereof. Between the one end **62c** and the other end **62d** of the second tube portion **62**, there are the exposed core wires **10e**, **52a**, **53a**, **54a**, a fusion spliced portion of branch optical fiber cord **10d**, and optical fiber cords **52** to **54**, and a sleeve **59** that protects the fusion spliced portion.

(48) A plurality of projection **62f** located between the one end **62c** and the other end **62d** of the second tube portion **62** in the first direction X are provided on the inner wall **62e** of the second tube portion **62**. The projections **62f** are ribs projecting toward a center axis of the second tube portion **62**, and have elasticity. Each the projection **62f** is regarded as a part of the second tube portion **62**. Therefore, the region where the projections **62f** are provided in the second tube portion **62** is a reduced diameter region whose inner diameter is smaller than that of the other region. The inner diameter of the reduced diameter region is the same or substantially the same as the outer diameter of the sleeve **59**. The projections **62f** are provided side by side along the first direction X, for example. In addition, the projections **62f** are provided side by side along the circumferential direction of the inner wall **62e** when viewed from the first direction X, but is not limited thereto.

Viewed from the first direction X, the projections **62f** may have an annular shape. From the viewpoint of preventing breakage of the branched optical fiber cable **15-1** accommodated in the second tube portion **62**, the distal end of the projection **62f** is rounded.

(49) An outer face **59a** of the sleeve **59** is sandwiched by projections **62f** each of which is a part of the second tube portion **62**. Accordingly, the movement of the sleeve **59** in the second tube portion **62** is regulated by the projections **62f**. In the present embodiment, both one end portion and the other end portion of the sleeve **59** in the first direction X are sandwiched by the projections **62f**.

(50) The partition wall **63** is a member that partitions the first tube portion **61** and the second tube portion **62**, and extends from the one end **62c** of the second tube portion **62** to the other end **62d** of the second tube portion **62** in the first direction X. Therefore, the interior space of the first tube portion **61** and the interior space of the main tube portion **71** included in the second tube portion **62** are separated from each other. A part of the partition wall **63** functions as a part of an inner wall of the first tube portion **61**, and another part of the partition wall **63** functions as a part of the inner wall **62e** of the second tube portion **62**. In the present embodiment, one end of the partition wall **63** in the first direction X functions as a part of the one end **62c** of the second tube portion **62**, and the other end of the partition wall **63** in the first direction X functions as a part of the other end **61d** of the first tube portion **61** in addition to a part of the other end **62d** of the second tube portion **62**.

(51) As shown in FIG. 6, a first thermal shrinkage tube **81** and a second thermal shrinkage tube **82** are mounted on the trunk cable **10** and the branch protection case **40-1**. In FIG. 6, cross sectional views of the first thermal shrinkage tube **81** and the second thermal shrinkage tube **82** are shown. Each of the first thermal shrinkage tube **81** and the second thermal shrinkage tube **82** is a resin member that fixes the position of the branch protection case **40-1** in the optical fiber cable **1**. In the present embodiment, each of the first thermal shrinkage tube **81** and the second thermal shrinkage tube **82** is a member (thermal shrinkage member) having a cylindrical shape and exhibiting a thermal shrinkage property. The thermal shrinkage member is a member having a property of shrinking, for example, by heating. From the viewpoint of wearability, a slit may be provided in each of the first thermal shrinkage tube **81** and the second thermal shrinkage tube **82**. In this case, each of the first thermal shrinkage tube **81** and the second thermal shrinkage tube **82** may have a C-shaped cross section.

(52) The first thermal shrinkage tube **81** is provided at a position surrounding the one end **61c** of the first tube portion **61**. The inner diameter of the first thermal shrinkage tube **81** may be equal to or larger than one end of the main body **60** in the first direction X. A portion **81a** of the first thermal shrinkage tube **81** extends away from the first tube portion **61** in the first direction X and surrounds a part of the trunk cable **10**. In other words, the portion **81a** is located on the opposite side of the other end **61d** through the one end **61c** of the first tube portion **61** in the first direction X. The second thermal shrinkage tube **82** is provided at a position surrounding the other end **61d** of the first tube portion **61** and the other end **62d** of the second tube portion **62**. The inner diameter of the second thermal shrinkage tube **82** may be equal to or larger than the other end of the main body **60** in the first direction X. A portion **82a** of the second thermal shrinkage tube **82** extends away from the first tube portion **61** in the first direction X and surrounds another portion of the trunk cable **10**. In other words, the portion **82a** is located on the opposite side of the one end **61c** through the other end **61d** of the first tube portion **61** in the first direction X.

(53) In the optical fiber cable **1**, each of the first thermal shrinkage tube **81** and the second thermal shrinkage tube **82** is in a thermally shrunk state. Thus, the first thermal shrinkage tube **81** is in close contact with the one end **61c** of the first tube portion **61** and a part of the trunk cable **10** located near the one end **61c**. The second thermal shrinkage tube **82** is in close contact with the other end **61d** of the first tube portion **61**, a part of the trunk cable **10** located near the other end **61d**, the other end **62d** of the second tube portion **62**, and a part of the branch optical fiber cable **15-1** located near the other end **62d**.

(54) According to the optical fiber cable **1** of the present embodiment described above, since the

partition wall **63** extending from the one end **62c** of the second tube portion **62** to the other end **62d** of the second tube portion **62** is provided, the trunk optical fiber cord **10c** and the branch optical fiber cord **10d** can be separated from each other in the branch protection case **40-1**. Accordingly, it is possible to suppress interference between the trunk optical fiber cord **10c** and the branch optical fiber cord **10d** in the branch protection case **40-1**. In addition, since the one end **62c** of the second tube portion **62** communicates with the interior space of the first tube portion **61**, the branch start position of the branch optical fiber cord **10d** can be covered by the branch protection case **40-1**. Accordingly, since the branch protection case **40-1** can satisfactorily protect the branch start position, it is possible to suppress damage to the branch optical fiber cord **10d** branched from the trunk optical fiber cord **10c**.

(55) In one example, the connecting portion is a fusion spliced portion of the core wire **10e** included in the branch optical fiber cord **10d** and the core wires **52a**, **53a** and **54a** included in the optical fiber cords **52** to **54**. Therefore, the fusion spliced portion can be favorably protected by the branch protection case **40-1**.

(56) In one example, the optical fiber cable **1** includes the reinforced tube **10b** covering the plurality of optical fiber cords **10a** and located at least outside of the branch protection case **40-1**, the reinforced tube **51a** covering optical fiber cords **52** to **54** and located at least outside of the branch protection case **40-1**, the first thermal shrinkage tube **81** as a resin member surrounding at least the one end **61c** of the first tube portion **61**, and the second thermal shrinkage tube **82** as a resin member surrounding at least the one end **61c** of the first tube portion **61**, the other end **61d** of the first tube portion **61**, and the other end **62d** of the second tube portion **62**, and the resin member is in close contact with at least a part of the branch protection case **40-1**, a part of the reinforced tube **10b**, and a part of the reinforced tube **51a**. In this case, rattling of the branch protection case **40-1** in the optical fiber cable **1** can be suppressed by the resin member.

(57) In one example, the optical fiber cable **1** includes a sleeve **59** covering the connecting portion, and an outer face **59a** of the sleeve **59** is sandwiched by the second tube portion **62**. Projections **62c** located between the one end **62c** and the other end **62d** of the second tube portion **62** are provided on the inner wall **62e** of the second tube portion **62**, and the projections **62f** are in close contact with the outer face **59a**. Therefore, since the movement of the connection portion in the second tube portion **62** can be suppressed, the connection portion is less likely to be damaged.

(58) In one example, the projections **62f** have elasticity. For this reason, it is possible to suppress damage to a part of the branched optical fiber cable **15-1** that is in contact with the projections **62f**.

(59) In one example, a first thermal shrinkage tube **81** surrounding the one end **61c** of the first tube portion **61** and a second thermal shrinkage tube **82** surrounding the other end **61d** of the first tube portion **61** and the other end **62d** of the second tube portion **62** are provided, the portion **81a** of the first thermal shrinkage tube **81** and the portion **82a** of the second thermal shrinkage tube **82** extend away from the first tube portion **61** in the first direction X, and the first thermal shrinkage tube **81** is in close contact with the one end **61c** of the first tube portion **61** and a part of the trunk cable **10** located near the one end **61c**. The second thermal shrinkage tube **82** is in close contact with the other end **61d** of the first tube portion **61**, a part of the trunk cable **10** located near the other end **61d**, the other end **62d** of the second tube portion **62**, and a part of the branch optical fiber cable **15-1** located near the other end **62d**. Therefore, rattling of the branch protection case **40-1** with respect to the optical fiber cable **1** can be suppressed by the first thermal shrinkage tube **81** and the second thermal shrinkage tube **82**.

(60) In one example, the branch protection case **40-1** includes a main body **60** having a first tube portion **61**, a second tube portion **62**, and a partition wall **63**, and the main body **60** is an assembly structure of a first main body portion **60a** defining a first part **61a** of the first tube portion **61**, a first part **62a** of the second tube portion **62**, and a first part **63a** of the partition wall **63** and a second main body portion **60b** defining a second part **61b** of the first tube portion **61**, a second part **62b** of the second tube portion **62**, and a second part **63b** of the partition wall **63**. Therefore, it is easy to

accommodate the trunk cable **10** and the branch optical fiber cable **15-1** in the branch protection case **40-1**.

(61) In one example, the branch protection case **40-1** includes connecting portions **60c** and **60d** that connect the first main body portion **60a** and the second main body portion **60b**. Therefore, the first main body portion **60a** and the second main body portion **60b** can be easily assembled. In addition, it is possible to suppress loss of either the first main body portion **60a** or the second main body portion **60b**.

(62) The branch protection case and the optical fiber cable including the same according to the present disclosure are not limited to the above-described embodiments, and may be variously modified. For example, the first main body portion and the second main body portion included in the above-described embodiment are connected to each other by the connecting portion, but are not limited thereto. FIG. 7 is a schematic perspective view illustrating a branch protection case according to a modification. As shown in FIG. 7, the first main body portion **60a** and the second main body portion **60b** may be separate from each other.

(63) In the above-described embodiment, the optical fiber cable includes the first heat shrinkable tube and the second heat shrinkable tube as a resin member for one branch protection case, but is not limited thereto. For example, the resin member may not have a thermal shrinkage property. The optical fiber cable may include a resin member that surrounds at least one end of the first tube portion, the other end of the first tube portion, and the other end of the second tube portion. For example, the resin member for one branch protection case may be a single tube, tape, or the like. In the former case, for example, the resin member covers and adheres to the branch protection case and the reinforced tube of each cable. In the latter case, for example, the resin member may be a wound body wound around the branch protection case and the reinforced tube of each cable. Note that two or more tapes may be wound for one branch protection case. These wound bodies and tapes may have a thermal shrinkage property. Even in these cases, the rattling of the branch protection case with respect to the optical fiber cable is suppressed.

(64) In the above-described embodiment, the branch trunk cable is fusion-spliced to the second optical fiber cord, but is not limited thereto. For example, the branch trunk cable and the second optical fiber cord may be connected to each other by a connector or the like. In any case, the optical connecting portion between the branch trunk cable and the second optical fiber cord is accommodated in the second tube portion.

(65) Although the projections integrated with the inner wall of the second tube portion are provided in the above-described embodiment, the present disclosure is not limited thereto. The projections may be separate from the second tube portion. In this case, the projections are fixed on the inner wall of the second tube portion via an adhesive or the like.

Claims

1. An optical fiber cable comprising: a plurality of first optical fiber cords including a trunk optical fiber cord and a branch optical fiber cord; a second optical fiber cord connected to the branch optical fiber cord; and a case covering a part of the plurality of first optical fiber cords and a part of the second optical fiber cord, wherein the case includes: a first tube portion through which the trunk optical fiber cord is passed; a second tube portion adjacent to the first tube portion and through which the branch optical fiber cord is passed; and a partition wall partitioning the first tube portion and the second tube portion, wherein an optical connecting portion of the branch optical fiber cord and the second optical fiber cord is accommodated in the second tube portion, wherein a first end of the second tube portion is located between a first end of the first tube portion and a second end of the first tube portion, the first end of the second tube portion communicating with an interior space of the first tube portion, and wherein the partition wall extends from the first end of the second tube portion to a second end of the second tube portion.

2. The optical fiber cable according to claim 1, further comprising: a first tube located at least outside the case, the first tube covering the plurality of first optical fiber cords; a second tube located at least outside the case, the second tube covering the second optical fiber cord; and a resin member surrounding at least the first end of the first tube portion, the second end of the first tube portion, and the second end of the second tube portion, wherein the resin member is in contact with at least a part of the case, a part of the first tube, and a part of the second tube.
 3. The optical fiber cable according to claim 2, wherein the resin member has a thermal shrinkage property.
 4. The optical fiber cable according to claim 1, wherein the case includes a body including the first tube portion, the second tube portion, and the partition wall, and wherein the body is an assembly structure of a first main body portion and a second main body portion, the first main body portion defining a first part of the first tube portion, a first part of the second tube portion, and a first part of the partition wall, and the second main body portion defining a second part of the first tube portion, a second part of the second tube portion, and a second part of the partition wall.
 5. The optical fiber cable according to claim 4, further comprising a connecting portion connecting the first main body portion and the second main body portion.
 6. The optical fiber cable according to claim 1, further comprising a sleeve covering the optical connecting portion, wherein an outer face of the sleeve is sandwiched by the second tube portion.
 7. The optical fiber cable according to claim 6, wherein a projection located between the first end and the second end of the second tube portion is provided on an inner wall of the second tube portion, and wherein the projection is in contact with the outer face.
 8. The optical fiber cable according to claim 7, wherein the projection has elasticity.
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